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Hierarchical decision-making for UAVs' game via LLM enhanced multi-agent reinforcement learning

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ABSTRACT

For the multi-UAV game missions, intelligent agents face challenges in high-dimensional state processing, dynamic situational awareness, and cooperative decision-making that conventional methods cannot address effectively. This paper proposes a novel Large Language Model-enhanced Multi-agent Deep Reinforcement Learning scheme (LLM-MATD3) with a hierarchical decision-making architecture comprising an environment modeling layer, LLM task planner, and task-enhanced MATD3 network. The scheme first employs a hierarchical task decomposition mechanism that transforms high-level game directives from LLMs into continuous embedding vectors through task embedding techniques. Then, the LLM task planner analyzes global game situations and assigns differentiated tasks to UAVs, overcoming the limitation of traditional MATD3 frameworks that rely solely on local observations. Simulation results show significant improvements in win rate, remaining rate, and collaborative efficiency for UAVs, providing a innovative reference approach for multi-agent systems in complex game environments.

1. Introduction

Recently, rapid advances in Unmanned Aerial Vehicle (UAV) technology have emerged in aerospace field, and the performances of autonomous flight control and collaborative decision-making in complex environments directly determine the systems' comprehensive capability [1]. Modern UAV systems face numerous technical challenges, with actuator fault and anti-disturbance control issues being particularly prominent, requiring intelligent tracking control strategies for flight safety [2]. Moreover, modern electromagnetic environments, characterized by strong electromagnetic pulses and intentional interference, severely impact UAVs' sensor modules and communication links [3]. In highly dynamic aerial game scenarios, external disturbances including gusts and turbulence continuously threaten flight stability, where integrating fixed-time observers with sliding mode control provides an effective solution [4]. However, the challenges of real-time cooperative decision-making, dynamic task allocation, and intelligent game strategies become critical problems when multiple UAVs work collaboratively in such complex game environments.

Multi-UAV cooperative systems enhance operational effectiveness through distributed architectures. For example, context-aware feature fusion [5] improves situational awareness but suffers from computational latency in dynamic environments; the asymptotically stable formation control [6] ensures convergence but exhibits slow transient responses, and intelligent task planning [7,8] excels in static scenarios but lacks real-time replanning capabilities. As agent numbers increase, the complexity of decision and control grows exponentially. In [9], Expert rule-based systems achieve high efficiency but limited generalization. The game-theoretic methods [10] provide rigorous frameworks but incur prohibitive computational costs for Nash equilibrium solutions. In addition, traditional optimization algorithms struggle with real-time constraints, high-dimensional spaces, and convergence in dynamic environments [11–13]. Position-weight schemes [14] lack stability under disturbances, while robust reinforcement learning [15] shows limited adaptability to unseen task scenarios. These bottlenecks hinder multi-UAV cooperative systems deployment.

With the advancement of artificial intelligence, Deep Reinforcement Learning (DRL) has been extensively applied to UAV decision-making [16], evolving from DQN [17] for discrete actions to DDPG [18] for continuous control and PPO's CTDE framework [19]. However, there are still some limitations: Wang et al. [20] combined SAC with HER for path planning but faced high-dimensional computational bur-

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den; Liu et al. [21] proposed a fixed-time ADP framework for multi-quadrotor coordination but primarily addressed formation tasks rather than game scenarios; TD3 [22] improved stability through twin Q-networks but struggled with multi-agent coordination. Multi-agent Reinforcement Learning now faces environmental non-stationarity, credit assignment, and dimensionality challenges [23]. Value decomposition methods like QMIX [24] and QTRAN [25] enable coordination but assume discrete action spaces. Recent advances show mixed results: it should be noted that some integrated results has been presented. For example, MAHPG [26] achieves policy diversity but scales poorly with agent count; ME-MADDPG [27] requires expert-designed potential fields, limiting adaptability; MATD3 [28] inherits TD3's stability but lacks global coordination in game scenarios. These constraints highlight the gap between theoretical advances and practical multi-UAV deployment requirements.

On the other hand, the breakthrough advances in Large Language Models (LLMs) offer novel solutions to address the mentioned limitations [29]. LLMs possess robust capabilities in contextual understanding, logical reasoning, and knowledge synthesis, enabling task comprehension from high-level semantic perspectives to global policy generation. Notably, LLMs offer advantages over rule-based methods through contextual reasoning and adaptive strategy generation in complex game scenarios. Recent studies demonstrate that integrating LLMs with reinforcement learning significantly enhances sample efficiency and generalization capabilities [30]. Zhu et al. proposed the LAMARL framework [31], which leverages LLMs to automatically generate prior policies and reward functions for MARL, achieving cooperative policy generation without manual design while significantly improving sample efficiency through prior policy integration. Liu et al. introduced the LGPF framework [32], utilizing LLMs' few-shot generalization to transform natural language descriptions into formation patterns, combined with CTDE architecture for precise swarm robotics control. However, it is also an open problem to effectively integrate LLMs' high-level cognitive capabilities with MARL's real-time decision-making abilities, particularly in highly dynamic and aerial game environments.

It should be noted that the existing approaches for multi-UAV game decision-making still suffer from three following limitations: (1) traditional RL frameworks (such as MATD3) rely solely on local observations without global situational awareness [33]; (2) a semantic gap exists between LLM-generated high-level directives and continuous policy learning in reinforcement learning [31,32]; (3) an effective integration mechanism between LLMs' cognitive reasoning and MARL's real-time decision-making remains absent [34]. To solve the above limitations, this work proposes a LLM-MATD3 scheme that constructs a hierarchical decision-making framework bridging global game planning and local execution through LLM-MARL integration. The main contributions include:

(1) A large language model-based game analysis and task planning framework is proposed to comprehensively analyze complex game situations and assign task types to each UAV, enabling hierarchical decision-making control from macro game perspectives to micro execution.
(2) A hierarchical task decomposition mechanism is proposed, whereby a specialized game reward function decomposes abstract task labels into state-dependent continuous subgoal completion scores. It enhances the semantic expressiveness of task embeddings, enabling the transformation from discrete task assignments to continuous behavioral guidance.
(3) A task-aware Actor-Critic network structure is proposed that integrates enhanced task embeddings into both policy and value networks. The actor network generates task-conditioned actions by combining state observations with task-specific guidance, while the critic network learns task-dependent value functions that account for task label transitions. It also enables an adaptive decision-making performance across diverse task types.

2. Preliminaries and problem formulations

2.1. UAV's motion model

The UAVs are assumed to maintain a fixed altitude while performing inertially coordinated maneuvers. Based on this assumption, the kinematic model for a single UAV is formulated as follows [35]:

$$\dot{\boldsymbol{p}}_i = \boldsymbol{V}_i = \begin{bmatrix} v_i \cos\theta_i \\ v_i \sin\theta_i \end{bmatrix}$$
$$\theta_i = \arctan 2(v_y^i, v_x^i) \tag{1}$$

where $\boldsymbol{p}_i = [x_i, y_i]^\top$ denotes the position vector of the ith UAV, and the heading angle $\theta_i \in (-\pi, \pi]$ is defined relative to the X-axis of the inertial coordinate system. The control forces generate accelerations according to $\dot{v}_x^i = F_x^i/m_{\text{uav}}$ and $\dot{v}_y^i = F_y^i/m_{\text{uav}}$, which subsequently update the UAV's velocity and position. The system's control input vector is defined as $\boldsymbol{u}_i = [F_x^i, F_y^i]^\top$, where the control forces are bounded by $F_x^i, F_y^i \in [F_{\min}, F_{\max}]$ to respect actuator limitations. In the multi-UAV cooperative system, $\boldsymbol{V}_i = (v_i \cos\theta_i, v_i \sin\theta_i)^\top$ and $\boldsymbol{V}_j = (v_j \cos\theta_j, v_j \sin\theta_j)^\top$ represent the velocity vectors of UAVs from the A-team and B-team respectively.

The Blue UAVs (A-team) consists of i UAVs, while the Black UAVs (B-team) consists of j UAVs. Each UAV is equipped with sensors to collect situational information, and the target for both teams is to maximize damage inflicted on the opposing force while minimizing their own losses. For a given task space $\Omega = \{(x, y) \mid x \in [x_{\min}, x_{\max}], y \in [y_{\min}, y_{\max}]\}$, the relative distance D_{ij} between UAVs from A-team and B-team is defined as follows:

$$D_{ij} = \|\boldsymbol{d}_{ij}\| = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \tag{2}$$

Then, we introduce the following two key geometric parameters:

(1) Target aspect angle φ_{ij} is defined as:

$$\varphi_{ij} = \arccos\left(\frac{\boldsymbol{d}_{ij} \cdot \boldsymbol{V}_i}{\|\boldsymbol{d}_{ij}\|\|\boldsymbol{V}_i\|}\right), \quad \varphi_{ij} \in [0, \pi] \tag{3}$$

which represents the angle between the velocity vector $\boldsymbol{V}_i$ of UAV i, and the relative position vector $\boldsymbol{d}_{ij}$ points toward UAV j.

(2) Target entry angle q_{ij} is defined as:

$$q_{ij} = \arccos\left(\frac{-\boldsymbol{d}_{ij} \cdot \boldsymbol{V}_j}{\|\boldsymbol{d}_{ij}\|\|\boldsymbol{V}_j\|}\right), \quad q_{ij} \in [0, \pi] \tag{4}$$

which represents the angle between the velocity vector $\boldsymbol{V}_j$ of UAV j, and the relative position vector $-\boldsymbol{d}_{ij}$ points toward UAV i.

As illustrated in Fig. 1, the task capability of UAVs is characterised by sector-shaped approach zones centred at each UAV and extending along their velocity vector directions. L_i and L_j represent the approach zone of A and B teams respectively. Effective execution of tasks can be conducted when UAVs enter these sector zones ($D_{ij} \le L_i$ or $D_{ij} \le L_j$).

2.2. UAV's sensor

The UAVs in both A team and B team use an 11-element sensor array to detect obstacles. Each sensor measures distance up to a maximum range $l_{\max}$. The sensors give values $l_i \in [0, l_{\max}]$, and $l_i = 0$ means that no obstacle is detected. The sensor array output is expressed by the following formula:

$$\boldsymbol{d}_{\text{obs}} = \left[l_1, l_2, l_i, \ldots, l_{11}\right], \quad l_i \in \left[0, l_{\max}\right] \tag{5}$$

2.3. Relative scenario classification

In a one-to-one UAV game, the geometric interaction model is illustrated in Fig. 2. This model establishes the relationships $\varphi_{ji} = \pi - q_{ji}$ and $q_{ji} = \pi - \varphi_{ij}$ between target aspect and entry angles. These parameters enable the classification of game scenarios into four fundamental configurations.

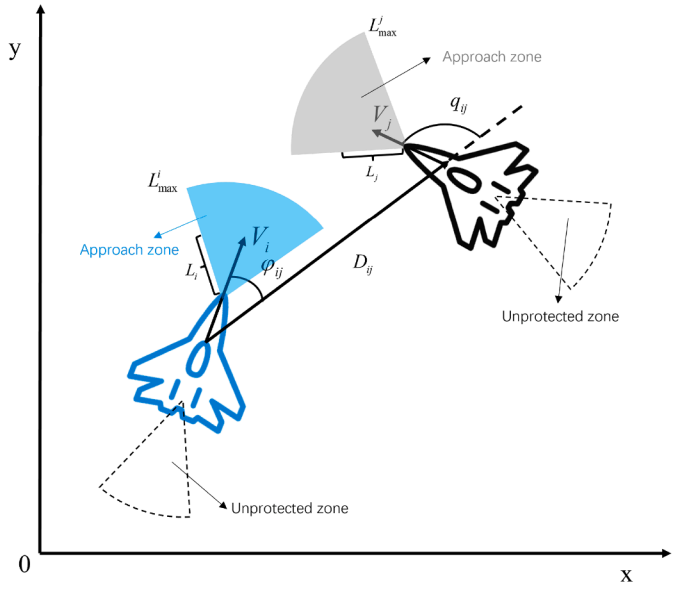


Fig. 1. UAV's model and approach zones.

(1) Balanced situation: When $\varphi_{ij} \geq \pi/2$ and $q_{ij} \geq \pi/2$, both UAVs are positioned in each other's rear area, with velocities either diverging or parallel.
(2) B-team advantageous situation: When $\varphi_{ij} < \pi/2$ and $q_{ij} < \pi/2$, the A-team is located in the B-team's forward area, and its velocity direction is directed away from the B-team.
(3) A-team advantageous situation: When $\varphi_{ij} \geq \pi/2$ and $q_{ij} < \pi/2$, the B-team is located in the A-team's forward area, and its velocity direction is deviating away from the A-team.
(4) Mutually disadvantageous situation: When $\varphi_{ij} < \pi/2$ and $q_{ij} \geq \pi/2$, both UAVs are located in each other's forward area, placing them in a state of mutual risk.

In the close-range game environment, the damage inflicted by each engagement on the opposing UAV is dynamic and limited. Each UAV is assumed to have an initial health H, when it suffers a loss in a task, it reduces its health by ΔH. When its health drops to 0, it is regarded as to be failed.

2.4. Markov game

The multi-UAV game system can be formalized as a Markov game, described by the tuple $(N, \boldsymbol{S}, \boldsymbol{A}, \boldsymbol{\Gamma}, R, \gamma)$. Specifically, N represents the number of UAVs in the system, and $\boldsymbol{S} = [\boldsymbol{S}_1, \ldots, \boldsymbol{S}_N]$ refers to the collective state space. Here, $\boldsymbol{S}_i$ represents the state space of the ith UAV, and $\boldsymbol{s}_i^t \in \boldsymbol{S}_i$ indicates the situational information observed by UAV i at time t. Correspondingly, $\boldsymbol{A} = [\boldsymbol{A}_1, \ldots, \boldsymbol{A}_N]$ is the joint action space, where $\boldsymbol{A}_i$ represents the set of executable actions for the ith UAV. Based on the observed states, each UAV adopts a stochastic policy $\pi_i : \boldsymbol{S}_i \times \boldsymbol{A}_i \to [0, 1]$ to select actions. The state transition function $\boldsymbol{\Gamma} : \boldsymbol{S} \times \boldsymbol{A}_1 \times \cdots \times \boldsymbol{A}_N \to \boldsymbol{S}'$ governs how the system evolves from one state to another based on joint actions. Upon executing an action, UAV i receives an immediate reward $r_i : \boldsymbol{S} \times \boldsymbol{A}_i \to \mathbb{R}$. The objective of each UAV is to maximize its expected cumulative reward: $R_i = \sum_{t=0}^{T} \gamma^t r_i^t$. The variable $\gamma \in [0, 1]$ denotes the discount factor that balances immediate and future rewards.

2.5. State and action space

In a multi-UAV game scenario, the A-team is the defender with its action decision-making trained using the proposed LLM-enhanced MATD3 scheme, which will be presented in the next section. For UAV i, its state space $\boldsymbol{S}_i$ consists of three components: instantaneous state information $\boldsymbol{S}_{\text{uav}}^i$, task information $\boldsymbol{S}_{\text{task}}^i$, and detection information $\boldsymbol{S}_{\text{detect}}^i$. Accordingly, the state space can be represented in the form of:

$$\boldsymbol{S}^i = \left[\boldsymbol{S}_{\text{uav}}^i, \boldsymbol{S}_{\text{task}}^i, \boldsymbol{S}_{\text{detect}}^i\right] \tag{6}$$

By defining two maximum position boundary parameters $x_{\max}$, $y_{\max}$, and the maximum velocity constraint $v_{\max}$, the instantaneous state information $\boldsymbol{S}_{\text{uav}}^i$ is normalized as follows:

$$\boldsymbol{S}_{\text{uav}}^i = \left[\frac{x_i}{x_{\max}}, \frac{y_i}{y_{\max}}, \frac{v_x^i}{v_{\max}}, \frac{v_y^i}{v_{\max}}, H_i\right] \tag{7}$$

where (x_i, y_i) denotes the spatial position coordinates of the UAV, and (v_x^i, v_y^i) represents its velocity components in the Cartesian coordinate system. Additionally, H_i denotes the UAV's current health status.

During task execution, the relative position information of the target needs to be integrated into the state space, as shown in the form of:

$$\boldsymbol{S}_{\text{task}}^i = \left[D_{ij}, \varphi_{ij}, q_{ij}\right] \tag{8}$$

To ensure formation stability and obstacle avoidance capability during the game, the detection information is integrated into the state space:

$$\boldsymbol{S}_{\text{detect}}^i = \left[\frac{l_1}{l_{\max}}, \frac{l_2}{l_{\max}}, \frac{l_3}{l_{\max}}, \ldots, \frac{l_{11}}{l_{\max}}\right] \tag{9}$$

where l_1, l_2, …, l_{11} represent the distances between the UAV and detected targets in different directions.

The action space of each UAV is defined as:

$$\boldsymbol{A}_i = \left[F_x^i, F_y^i\right]^\top \tag{10}$$

where F_x^i and F_y^i represent the control forces along the x-axis and y-axis respectively.

3. LLM-enhanced MATD3 framework and methods

In view of the existing LLM-enhanced reinforcement learning methods, they typically provide one-time guidance during pre-training or cold-start phases, such as LLM-guided exploration [34], prior policy generation [36], formation pattern description [37], and action distribution guidance [38]. Compared with the above results, the proposed LLM-MATD3 continuously integrates LLM instructions into the Actor-Critic network through the hierarchical task embedding mechanism, enabling real-time policy modulation across multi-agent coordination, continuous action spaces, and dynamic task allocation in game scenarios. In this section, we will present a novel LLM-enhanced MATD3 framework that integrates the game planning capabilities of large language models with the autonomous decision-making of deep reinforcement learning. As shown in Fig. 3, the framework comprises four components: the environment modeling layer, the LLM task planner, the task embedding network, and the task-enhanced MATD3 network. The LLM task planner analyzes situational dynamics and assigns specific tasks to each agent. Subsequently, the hierarchical task decomposition mechanism, guided by a specialized task game reward function, transforms abstract task labels into enhanced embedding vectors. These embeddings provide semantic guidance for reinforcement learning-based action generation.

3.1. Reward function

To guide effective agent behavior in a multi-UAV game, we design a hierarchical reward structure comprising components for task execution, safety enforcement, team coordination, and terminal evaluation.

(1) Task reward R_{task} assesses task execution performance through three components.
   (i) The angular advantage reward r_{angle} assesses the geometric relationship based on the aspect angles φ_{ij} and target angles q_{ij}.

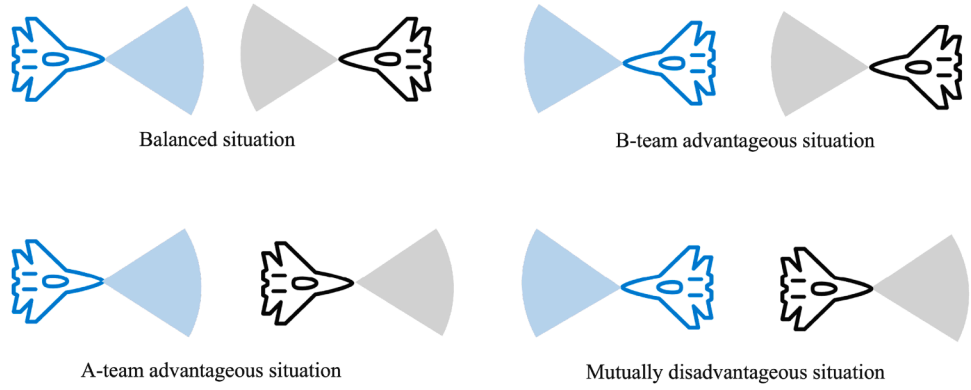


Fig. 2. 1 vs 1 scenario.

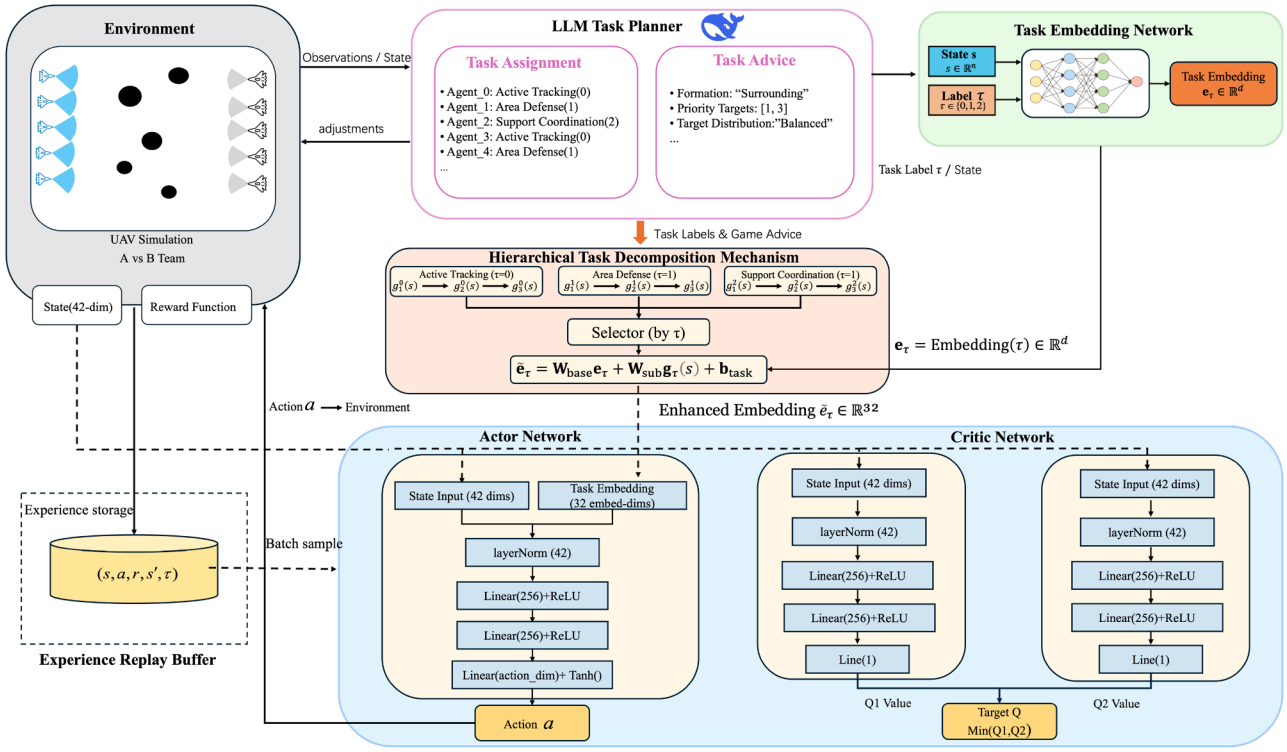


Fig. 3. LLM-enhanced MATD3 framework for multi-UAV coordinated decision making.

It is defined as:

$$r_{\text{angle}} = \begin{cases} 5.0, & \text{if } |\varphi_{ij}| < \pi/4 \text{ and } |q_{ij}| > 3\pi/4 \\ 2.5, & \text{if } |\varphi_{ij}| < \pi/2 \text{ and } |q_{ij}| > \pi/2 \\ 0.5, & \text{otherwise} \end{cases} \tag{11}$$

(ii) The distance reward r_{distance} encourages UAVs to maintain optimal engagement distance and it is formulated as:

$$r_{\text{distance}} = \begin{cases} \frac{D_{ij}}{d_{\min}} - 1, & \text{if } D_{ij} < d_{\min} \\ 1.5(1 - \frac{|D_{ij} - d_{\text{id}}|}{d_{\max} - d_{\min}}), & \text{if } d_{\min} \le D_{ij} \le d_{\max} \\ 1 - \min(1, \frac{D_{ij} - d_{\max}}{\rho_i \cdot 1.2}), & \text{if } D_{ij} > d_{\max} \end{cases} \tag{12}$$

where ρ_i denotes the task radius. The parameters $d_{\text{id}} = 0.7\rho_i$, $d_{\min} = 0.3\rho_i$, and $d_{\max} = 0.9\rho_i$ represent the ideal distance, lower bound, and upper bound of the effective task zone respectively.

(iii) The approach reward r_{approach} encourages UAVs to approach and engage the nearest target UAV:

$$r_{\text{approach}} = 8.0 \cdot \left(1.0 - \left(\frac{D_{ij}}{\rho_i \cdot 4.0}\right)^b\right) \tag{13}$$

Thus, the total task reward combines the above three components, and it is presented in the form of:

$$R_{\text{task}} = r_{\text{angle}} + r_{\text{distance}} + r_{\text{approach}} \tag{14}$$

(2) The safety reward R_{safety} penalizes environmental collision, UAV collision, and boundary penalties. This reward is formulated as:

$$R_{\text{safety}} = \begin{cases} -30, & d < 2 \text{ or } D_{ij} < 2 \\ -10 \cdot \frac{d_{\text{safe}} - d}{d_{\text{safe}}^2}, & 5 < d < 15 \\ -\exp[2(1 - \frac{d_{\text{border}}}{20})], & d_{\text{border}} < 20 \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where d denotes the distance to the nearest obstacle or UAV (in meters); d_{border} represents the minimum distance to mission boundaries; and d_{safe} defines the safety warning threshold.

(3) The team cooperation reward encourages numerical advantage and coordinated engagement, formulated as:

$$R_{\text{team}} = 10.0 \cdot \frac{5 - N_j}{5} + 1.5 \cdot N_{\text{task}} \tag{16}$$

where N_j denotes the remaining B-team UAVs, and N_{task} represents the number of A-team UAVs coordinating on the B-team.

(4) The terminal reward R_{terminal} is provided at the conclusion of each episode based on the game outcome, which is expressed in the form of:

$$R_{\text{terminal}} = \begin{cases} 20.0, & \text{if } N_j = 0 \text{ and } N_i > 0 \\ 10.0, & \text{if } N_i > N_j \text{ and } N_j > 0 \\ -10.0, & \text{if } N_i = 0 \\ 0.3 + 0.7 \cdot \frac{5-N_j}{5}, & \text{if } N_i > N_j \\ 0, & \text{otherwise} \end{cases} \tag{17}$$

where its structure implements a hierarchical objective system: complete mission success, maintaining operational advantage, performance assessment, and mission failure avoidance.

By analyzing the characteristics of UAV tasks in game missions, we propose a structured comprehensive reward function:

$$\begin{aligned} R_{\text{total}} &= \omega_1 \cdot R_{\text{task}} + \omega_2 \cdot R_{\text{safety}} \\ &+ \omega_3 \cdot R_{\text{team}} + \omega_4 \cdot R_{\text{terminal}} \end{aligned} \tag{18}$$

3.2. LLM-based task analysis and assignment

3.2.1. Task analysis

In traditional MATD3 frameworks [33], agents can only make decisions based on local observations, lacking the capability to comprehend the overall game situation. To address this issue, we introduces an LLM as a game analyzer to achieve global game awareness capability.

The game situation assessment function can be represented as:

$$\mathcal{E}(S) = \Phi_{\text{LLM}}(\boldsymbol{\Omega}_t, \boldsymbol{M}_t) \tag{19}$$

where Φ_{LLM} represents the analysis function of the LLM; $\boldsymbol{\Omega}_t$ denotes the current game state; and $\boldsymbol{M}_t$ is the historical game record. This function maps game data to situational assessment results.

The global game state $\boldsymbol{\Omega}_t$ contains complete state information for all UAVs, and it is formulated as follows:

$$\boldsymbol{\Omega}_t = \{\boldsymbol{\omega}_{\text{A}}^1, \ldots, \boldsymbol{\omega}_{\text{A}}^i, \boldsymbol{\omega}_{\text{B}}^1, \ldots, \boldsymbol{\omega}_{\text{B}}^j\} \tag{20}$$

where each UAV's state vector $\boldsymbol{\omega} = [p_x, p_y, v_x, v_y, \theta, H]$ includes position, velocity, heading, and health status.

The historical performance record stores game outcomes to capture task execution patterns and game effects. Each record contains the following information:

$$\boldsymbol{M}_i = [win_i, N_i, N_j, step] \tag{21}$$

where win_i indicates whether the ith A-team UAV was victorious (1 for victory, 0 for failure); N_i and N_j denote the number of the remaining A-team and B-team UAVs respectively; and $step$ represents the total number of time steps in the game episode.

The LLM task analysis evaluates situational parameters to generate the game assessment $\mathcal{E}(S)$, comprising formation recommendations A_{form} and target priorities $\boldsymbol{T}_{\text{priority}} = [\text{id:1}, \text{id:2}, \ldots, \text{id:7}]$. The key parameters are defined as follows:

$$r_{\text{num}} = \frac{N_i}{N_j} \tag{22}$$

$$d_{\text{A}} = \frac{1}{N_i(N_i - 1)} \sum_{i=1}^{N_i} \sum_{j \neq i}^{N_i} \|p_i - p_j\| \tag{23}$$

$$d_{\text{B}} = \frac{1}{N_j(N_j - 1)} \sum_{i=1}^{N_j} \sum_{j \neq i}^{N_j} \|p_i - p_j\| \tag{24}$$

where r_{num} represents the numerical ratio between teams; d_{A} and d_{B} quantify the spatial dispersion of A-team and B-team UAVs respectively.

Based on these metrics, the system recommends three formation tasks, as illustrated in Fig. 4.

(1) Concentrated Formation: This formation is deployed when $r_{\text{num}} < 0.8$ or $d_{\text{B}} > 2d_{\text{A}}$, forming triangular patterns to maximize approach density.

(2) Scattered Formation: This formation is deployed when $0.8 \leq r_{\text{num}} \leq 1.3$, assigning individual operational zones to maximize coverage while minimizing exposure to B-team UAVs.

(3) Surrounding Formation: This formation is deployed when $r_{\text{num}} > 1.3$ or $d_{\text{B}} < 0.5d_{\text{A}}$, enabling encirclement maneuvers against concentrated B-team UAVs.

3.2.2. Task assignment

Based on the game analysis $\mathcal{E}(S_t)$, the LLM uses a mapping function $\mathcal{T} : \mathcal{E}(S_t) \times \boldsymbol{M}_t \to \{0, 1, 2\}^n$ to assign each UAV to one of three task types, as shown in Fig. 5. Each task type corresponds to specific game behaviors.

(1) Active Tracking ($\tau = 0$): The UAV is responsible for actively approaching UAVs with the nearest one. This involves maneuvering to gain positional advantage, especially when the target is within the favorable engagement zone (where $|\varphi_{ij}| < \pi/4$ and $|q_{ij}| > 3\pi/4$). The ideal Tracking distance is maintained at approximately $d_{\text{id}} = 0.7\rho_i$.

(2) Area Defense ($\tau = 1$): The UAV is responsible for key areas of the game, preventing B-team UAVs from approaching. Through boundary-constraint mechanisms (r_{boundary}) that limit the operational range, these UAVs can actively track B-team UAVs approaching the key zone without pursuing targets outside their assigned area.

(3) Support Coordination ($\tau = 2$): The UAV is responsible for enhancing overall game effectiveness through team coordination. UAVs assigned to this task maintain formation integrity by preserving optimal team distance (d_{team}) to strengthen numerical advantage.

The LLM-enhanced MATD3 framework enables comprehensive analysis of complex game situations and task assignments. Fig. 6 illustrates the LLM-based assignment process, showing how the module analyzes the game situation from input data and delivers the assignment results in JSON format.

3.3. Hierarchical task decomposition mechanism

To bridge the gap between discrete task labels and continuous policy learning, this subsection proposes a hierarchical task decomposition mechanism that augments task embeddings with state-dependent sub-goal information. This mechanism serves as an intermediate layer between the LLM task planner and the Actor-Critic network, transforming abstract task labels $\tau \in \{0, 1, 2\}$ into task completion features. For each task type, as illustrated in Fig. 5, we define a sub-goal set $\mathcal{G}_\tau = \{g_1^\tau, g_2^\tau, g_3^\tau\}$, where each sub-goal function $g_k^\tau : \mathcal{S} \to [0, 1]$ evaluates the completion degree of a specific sub-objective.

(1) Active Tracking Task ($\tau = 0$) Decomposition:

$$\begin{cases} g_1^0(s) = \max\limits_{j=1}^{N_j} \left[0.6 \cdot \max\left(0, 1 - \dfrac{D_{ij}}{1.5\rho_i}\right) + 0.4 \cdot \dfrac{H_j}{H}\right] \\ g_2^0(s) = \exp\left(-\dfrac{(D_{ij^*} - 0.7\rho_i)^2}{0.1\rho_i^2}\right) \cdot \max\left(0, 1 - \dfrac{|\varphi_{ij^*}|}{\pi/4}\right) \end{cases} \tag{25}$$

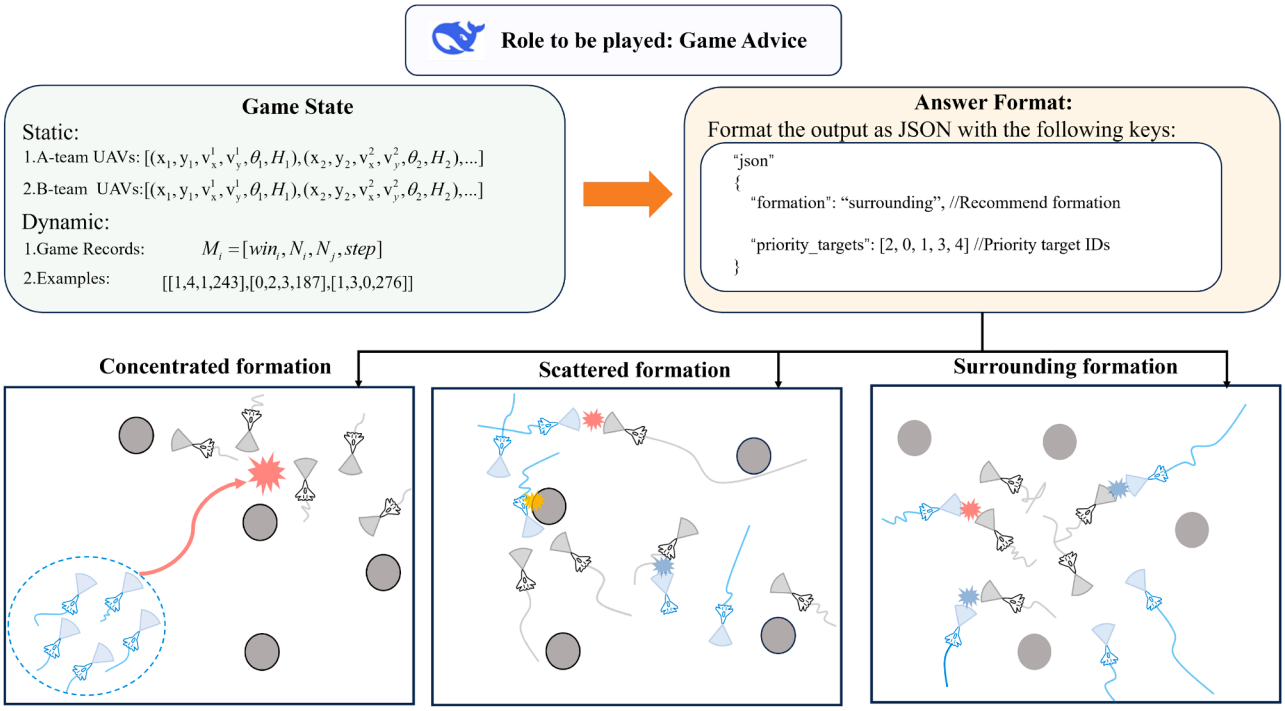


Fig. 4. LLM-enhanced MATD3 algorithm for UAV formation allocation.

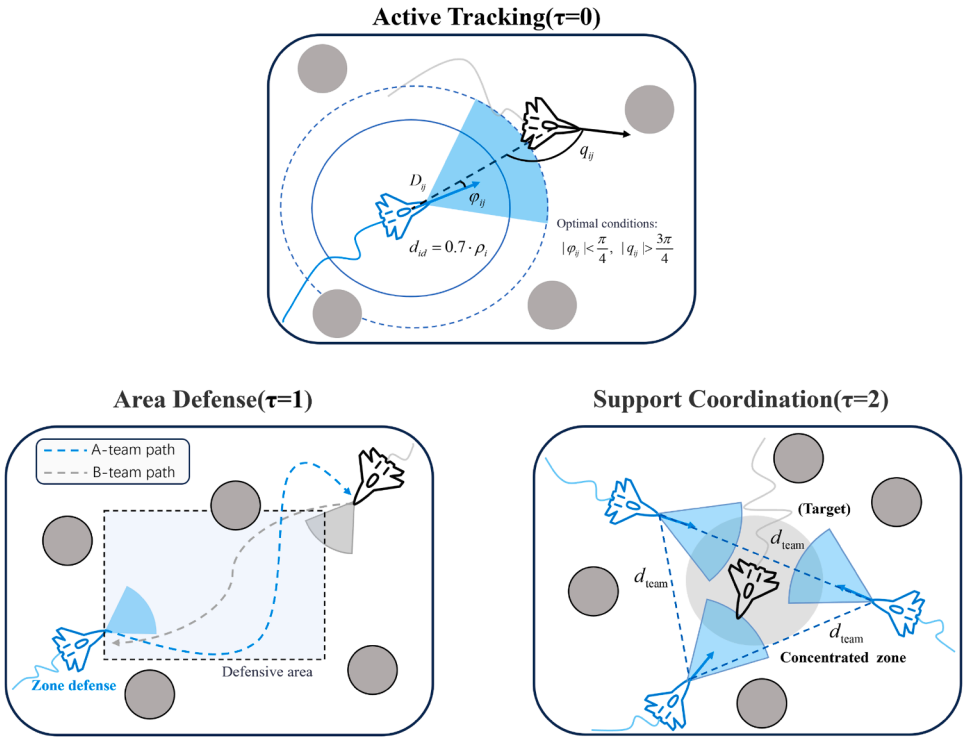


Fig. 5. UAV Task Labels.

$$\begin{cases} g_3^0(s) = \left(1 - \dfrac{D_{ij^*}}{\rho_i}\right)\left(1 - \dfrac{|\varphi_{ij^*}|}{\pi/4}\right), & \text{if } D_{ij^*} \le \rho_i \text{ and } |\varphi_{ij^*}| < \dfrac{\pi}{4} \\ g_3^0(s) = 0, & \text{otherwise} \end{cases} \tag{26}$$

where j^* denotes the target index selected by g_1^0. Specifically, $g_1^0(s)$ represents target selection based on distance and health status; $g_2^0(s)$ evaluates position optimization towards ideal distance $0.7\rho_i$ and angle $|\varphi_{ij^*}| < \pi/4$; $g_3^0(s)$ measures task execution effectiveness, activated only when task conditions are satisfied.

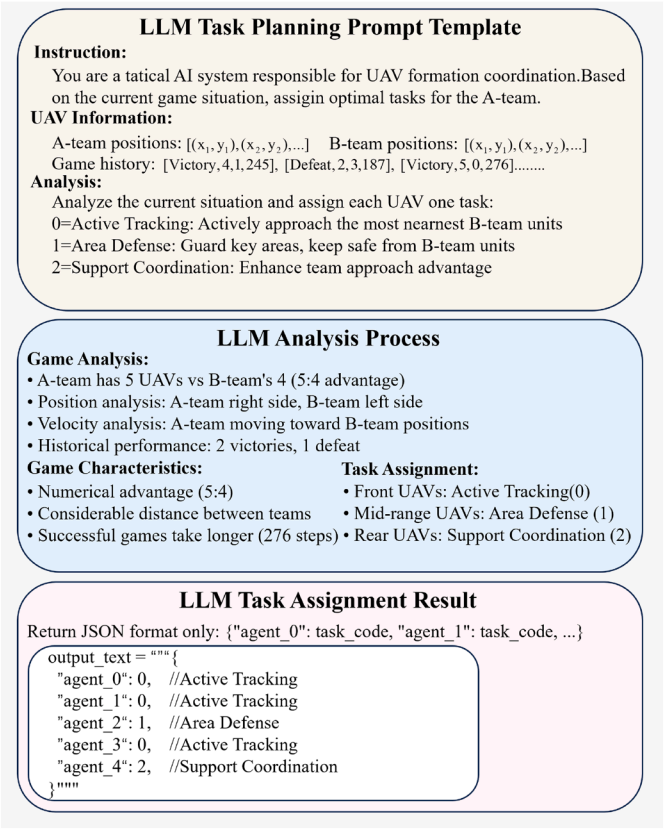


Fig. 6. LLM task assignment process.

(2) Area Defense Task ($\tau = 1$) Decomposition:

$$\begin{cases} g_1^1(s) = \exp\left(-\dfrac{d_{\text{border}}^2}{0.2\rho_i^2}\right) \\ g_2^1(s) = \max\left(0, 1 - \dfrac{d_{\text{border}}}{2\rho_i}\right) \\ g_3^1(s) = \dfrac{1}{N_j}\sum_{j=1}^{N_j} \exp\left(-\dfrac{D_{ij}^2}{2\rho_i^2}\right) \end{cases} \tag{27}$$

where $g_1^1(s)$ measures positioning accuracy within the defense zone; $g_2^1(s)$ evaluates proximity to the assigned defense boundary; $g_3^1(s)$ assesses tracking capability based on distance to B-team UAVs.

(3) Support Coordination Task ($\tau = 2$) Decomposition:

$$\begin{cases} g_1^2(s) = \exp\left(-\dfrac{(d_{\text{team}} - \rho_i)^2}{0.5\rho_i^2}\right) \\ g_2^2(s) = \dfrac{N_{\text{task}}}{N_i} \\ g_3^2(s) = \dfrac{1}{N_i - 1}\sum_{k\neq i}^{N_i} \max\left(0, 1 - \dfrac{D_{ik}}{\rho_i}\right) \end{cases} \tag{28}$$

where $g_1^2(s)$ measures formation coherence around ideal distance ρ_i; $g_2^2(s)$ evaluates the coordinated task ratio; $g_3^2(s)$ assesses mutual support capability among teammates within effective range.

The hierarchical task decomposition mechanism constructs enhanced embedding vectors based on sub-goal completion scores. The three sub-goal completion scores are first combined into a vector $\boldsymbol{g}_\tau(s) = [g_1^\tau(s), g_2^\tau(s), g_3^\tau(s)]^\top$, and then integrated with the original task embeddings through a linear transformation as follows:

$$\widetilde{\boldsymbol{e}}_\tau = \boldsymbol{W}_{\text{base}}\boldsymbol{e}_\tau + \boldsymbol{W}_{\text{sub}}\boldsymbol{g}_\tau(s) + \boldsymbol{b}_{\text{task}} \tag{29}$$

where $\boldsymbol{W}_{\text{base}} \in \mathbb{R}^{d\times d}$ transforms the original task embedding, $\boldsymbol{W}_{\text{sub}} \in \mathbb{R}^{d\times 3}$ projects sub-goal completion scores into the embedding space, and $\boldsymbol{b}_{\text{task}} \in \mathbb{R}^d$ provides task-specific bias.

Remark 1. The sub-objective functions (25)–(28) are designed by the game principles and mathematical considerations. For the Active Tracking ($\tau = 0$) task, g_1^0 uses a 6:4 distance-health weighting to prioritize positional advantage; g_2^0 employs a Gaussian kernel centered at the optimal engagement distance $0.7\rho_i$; and g_3^0 implements threshold activation to prevent false positives. For Area Defense ($\tau = 1$), exponential decay in g_1^1 ensures precise zone positioning, while linear guidance in g_2^1 maintains long-range stability. For Support Coordination ($\tau = 2$), g_1^2 is designed to improve the safety of the formation, and the Gaussian function-based distances constraints of each UAVs are limited within $[0.5\rho_i, 1.5\rho_i]$. These functions are systematically derived from UAV physical characteristics, geometric constraints in game scenarios, and multi-agent coordination requirements.

3.4. Task-aware actor-critic network

In this subsection, we improve the traditional MATD3 RL frameworks [33]. Especially, the Actor-Critic networks will be redesigned by introducing task embedding network layer. The proposed task-aware architecture incorporates a task embedding layer that integrates discrete task information into the policy network, enabling agents to adjust decision-making behaviors based on task types. The task embedding layer transforms discrete task labels into continuous vector representations, formalized as:

$$\boldsymbol{e}_\tau = \text{Embedding}(\tau) \in \mathbb{R}^d \tag{30}$$

where $\text{Embedding}(\cdot)$ is the mapping function parameterized by embedding matrix $\boldsymbol{E} \in \mathbb{R}^{|T|\times d}$, and d is the embedding dimension. Note that $\boldsymbol{e}_\tau$ represents the original task embedding, which is subsequently enhanced through the hierarchical task decomposition mechanism to generate $\widetilde{\boldsymbol{e}}_\tau$.

The actor network can be represented as a mapping $\pi_\theta : \boldsymbol{S} \to \boldsymbol{A}$, where $\boldsymbol{S}$ is the state space and $\boldsymbol{A}$ is the action space. This research extends the standard actor network by introducing the hierarchical task decomposition mechanism, forming a task-aware actor network as follows:

$$a = \pi_\theta(s, \widetilde{\boldsymbol{e}}_\tau) = \tanh(\text{MLP}(s, \lambda_\tau \cdot \widetilde{\boldsymbol{e}}_\tau)) \tag{31}$$

where MLP represents a multi-layer perceptron, and λ_τ is a weighting parameter controlling the influence of task embedding on action generation.

The task-aware critic network learns the state-action-task value function $Q(s, a, \widetilde{\boldsymbol{e}}_\tau)$, which represents the expected cumulative reward of executing action a under task τ. The formulation is given by:

$$Q(s, a, \widetilde{\boldsymbol{e}}_\tau) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s = s_0, a = a_0, \tau = \tau_0\right] \tag{32}$$

TD3 employs a twin-critic architecture to mitigate Q-value overestimation and uses target networks to improve training stability. The target Q-value calculation takes into account task label transitions as follows:

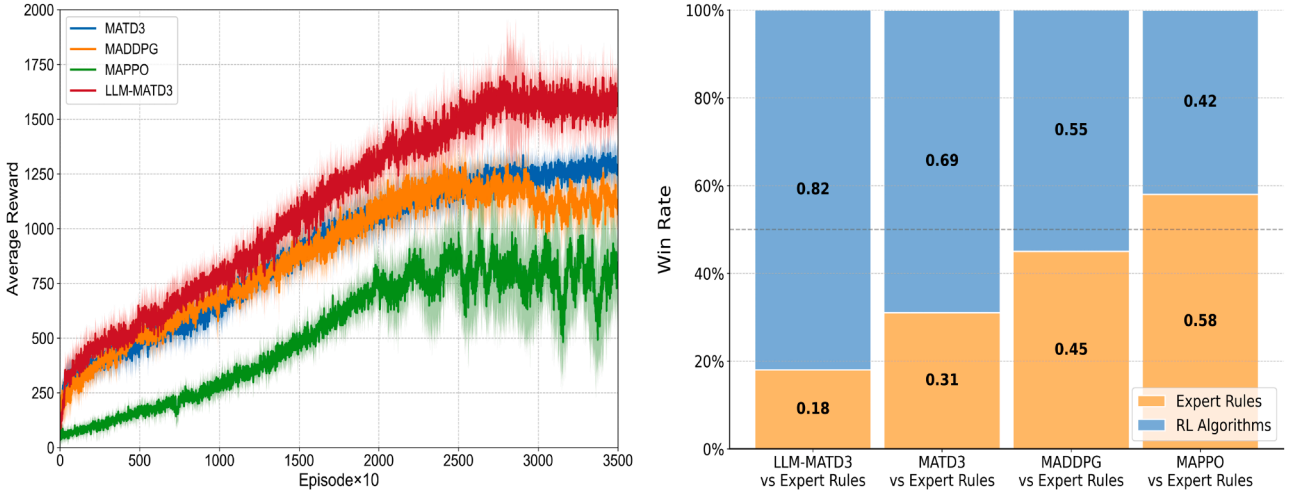
$$y = r + \gamma \min_{i=1,2} Q_{\phi_i'}(s', \pi_{\theta'}(s', \widetilde{\boldsymbol{e}}_{\tau'}) + \epsilon, \widetilde{\boldsymbol{e}}_{\tau'}) \tag{33}$$

where τ' is the next-state task label, and $\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma^2), -c, c)$ is clipped noise for policy smoothing.

The critic network is optimized by minimizing the temporal difference error. Its loss function and gradient update are defined as follows:

$$\begin{cases} L(\phi_i) = \mathbb{E}_{(s,a,r,s',\tau)\sim D}\left[(Q_{\phi_i}(s, a, \widetilde{\boldsymbol{e}}_\tau) - y)^2\right] \\ \nabla_{\phi_i} L(\phi_i) = \mathbb{E}_{(s,a,r,s',\tau)\sim D}\left[2(Q_{\phi_i}(s, a, \widetilde{\boldsymbol{e}}_\tau) - y) \cdot \nabla_{\phi_i} Q_{\phi_i}(s, a, \widetilde{\boldsymbol{e}}_\tau)\right] \end{cases} \tag{34}$$

The actor network is optimized using the deterministic policy gradient algorithm. The objective function and gradient update



(a) Cumulative reward comparison of different algorithms

(b) Win rate comparison of different algorithms

Fig. 7. Performance comparison of different algorithms in UAV game environments.

Algorithm 1 LLM-MATD3 algorithm.

Initialize critic networks Q_{ϕ_1}, Q_{ϕ_2}, and actor network π_θ with parameters ϕ_1, ϕ_2, θ
Initialize target networks $\phi_1' \leftarrow \phi_1$, $\phi_2' \leftarrow \phi_2$, $\theta' \leftarrow \theta$
Initialize embedding matrix $E \in \mathbb{R}^{|T|\times d}$
Initialize replay buffer D
for each episode do **do**
  Initialize state s_1 and task labels $\tau = \{\}$
  for each timestep t do **do**
    if t mod task_interval $= 0$ then **then**
      $\mathcal{T} \leftarrow LLM$.assign_tasks$(P_i, P_j, N_i, N_j)$
    end if
    for each agent i do **do**
      $\tau_i \leftarrow \mathcal{T}$.get("agent_" $+ i, 0$)
      Compute sub-goal vector: $\mathbf{g}_{\tau_i}(s_i)$
      Generate enhanced embedding: $\widetilde{\boldsymbol{e}}_{\tau_i}$
      $a_i \leftarrow \pi_\theta(s_i, \widetilde{\boldsymbol{e}}_{\tau_i}) + \varepsilon$
    end for
    Execute a, observe r, s'
    Store transition tuple (s, a, r, s', τ) in D
    Sample mini-batch of N transitions $(s_i, a_i, r_i, s_i', \tau_i)$ from D
    Compute enhanced embeddings for next states: $\widetilde{\boldsymbol{e}}_{\tau'}$
    $\tilde{a}_i' \leftarrow \pi_{\theta'}(s_i', \widetilde{\boldsymbol{e}}_{\tau'}) + \varepsilon$
    $y_i \leftarrow r_i + \gamma \min_{j=1,2} Q_{\phi_j'}(s_i', \tilde{a}_i', \widetilde{\boldsymbol{e}}_{\tau'})$
    Update critics $\phi_i \leftarrow L = E_{(s,a,r,s',\tau)\sim D}[(Q_{\phi_i}(s, a, \widetilde{\boldsymbol{e}}_\tau) - y)^2]$
    if t mod $d_{\text{policy}} = 0$ **then**
      Update θ by the deterministic policy gradient: $\nabla_\theta J(\theta) = E[(\nabla_a Q_{\phi_i}(s, a, \widetilde{\boldsymbol{e}}_\tau))|_{a=\pi_\theta(s,\widetilde{\boldsymbol{e}}_\tau)} \cdot \nabla_\theta \pi_\theta(s, \widetilde{\boldsymbol{e}}_\tau)]$
      Update target networks:
      $\theta' \leftarrow \tau_u\theta + (1-\tau_u)\theta'$
      $\phi_i' \leftarrow \tau_u\phi_i + (1-\tau_u)\phi_i', i \in \{1, 2\}$
    end if
    $s \leftarrow s'$
  end for
  Record game results for LLM
end for

are defined as:

$$\begin{cases} J(\theta) = \mathbb{E}_{(s,\tau)\sim D}\left[Q_{\phi_i}(s, \pi_\theta(s, \widetilde{\boldsymbol{e}}_\tau), \widetilde{\boldsymbol{e}}_\tau)\right] \\ \nabla_\theta J(\theta) = \mathbb{E}_{(s,\tau)\sim D}\left[\nabla_a Q_{\phi_i}(s, a, \widetilde{\boldsymbol{e}}_\tau)|_{a=\pi_\theta(s,\widetilde{\boldsymbol{e}}_\tau)} \cdot \nabla_\theta \pi_\theta(s, \widetilde{\boldsymbol{e}}_\tau)\right] \end{cases} \tag{35}$$

To maintain training stability, TD3 employs soft updates for update the target network parameters:

$$\begin{cases} \theta' \leftarrow \tau_u\theta + (1-\tau_u)\theta' \\ \phi_i' \leftarrow \tau_u\phi_i + (1-\tau_u)\phi_i' \end{cases} \tag{36}$$

Summing up the above analysis and design, the Algorithm 1 illustrates the core-code of the proposed LLM-MATD3 in this work.

Remark 2. There are some existing paradigms for the interaction between RL and LLM, such as external rewards [39], critical-moment guidance [40], or regularization terms [41], or cold-start exploration guidance [34]. Unlike the above results, the proposed LLM-MATD3 mechanism in this work achieves direct modulation of the policy update direction through semantic embedding. In the proposed framework, the LLM's discrete task label τ is transformed into the enhanced task embedding $\tilde{e}_\tau$ through semantic encoding and state enhancement in (29)–(30), which directly enters the Actor-Critic computational graph. Since $\tilde{e}_\tau$ appears simultaneously in both multiplicative factors of the policy gradient (35), a change in the LLM's task assignment directly redirects $\nabla_\theta J(\theta)$.

4. Experiments and results

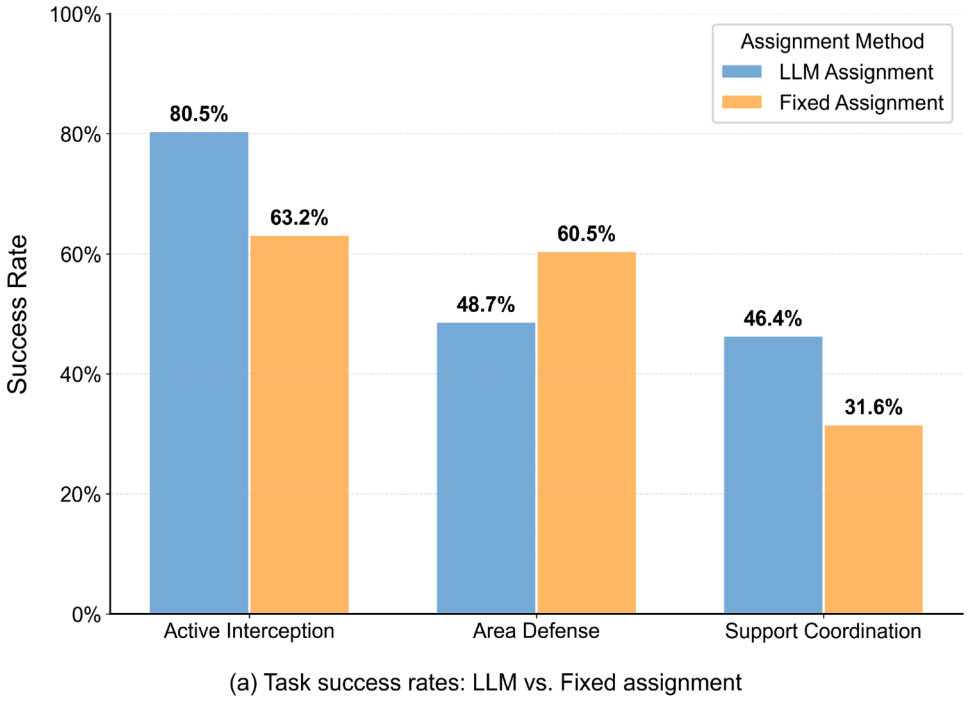
This section validates the proposed LLM-MATD3 framework through three complementary experiments: baseline comparisons, ablation studies on task allocation, and scalability analysis across varied task scales.

4.1. Experimental settings

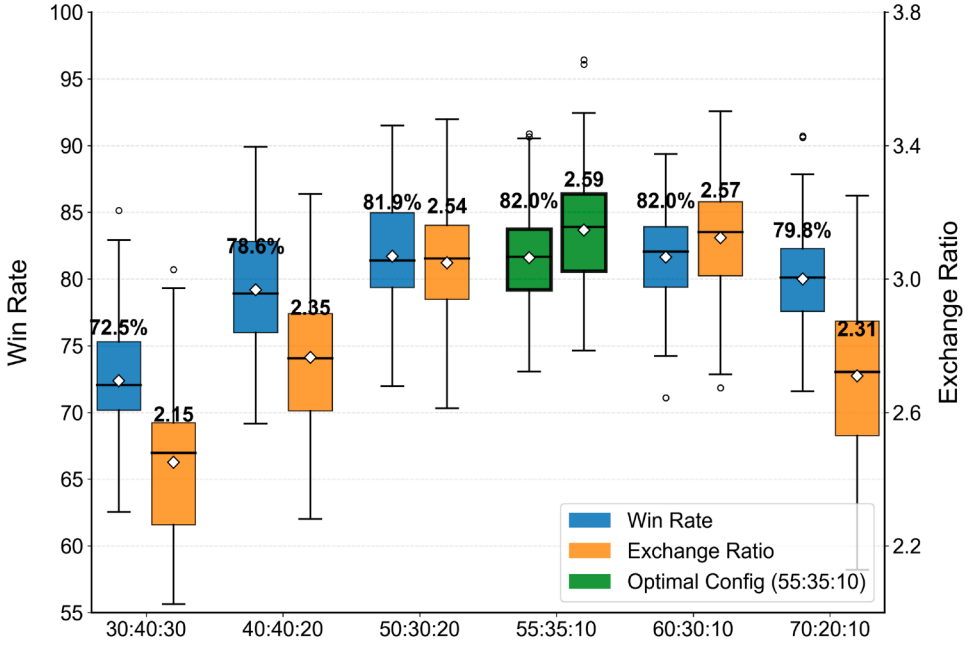
The environment comprises a 1600×1600 square-meter operational space with randomly distributed static obstacles, shown in Table 1. Team A and Team B each deploy 5 UAVs, with initial positions on the western and eastern boundaries respectively. Team A employs the proposed LLM-MATD3 scheme, while Team B adopts an expert knowledge-based system [9], which encodes experience to prioritize engaging the nearest detected UAVs. Each episode lasts up to 400 timesteps and terminates when all UAVs of one team are eliminated or the maximum timestep is reached. Each UAV is initialized with a health value of $H_0 = 100$ and is eliminated when its health reaches zero. The dynamic of UAV interactions, including calculation of health value and collision

handling, follow the physical model in Section 3. The training process adopts hyperparameters listed in Table 2, with random seeds used to ensure experimental reproducibility. Each training session runs for 35,000 episodes using Python 3.8 and PyTorch 1.9.0 as the underlying computational framework.

Regarding to the reasons and principles for key parameter settings, the task radius $\rho_i = 300.0$ m is set to 1.5 times the sensor detection range ($l_{\max} = 200.0$ m), ensuring UAVs can complete the full detection-tracking-execution process while serving as the normalization basis for the reward structure in (12). The safety distance threshold $d_{\text{safe}} = 15$ m is derived from kinematic constraints: given $dt = 0.1$s and $v_{\max} = 45$ m/s, and the maximum displacement during a 2–3 step response cycle is approximately 9–13.5 m, thus 15 m can provide sufficient safety margin. For the environment modeling rules, effective engagement is determined when $D_{ij} \le L_i$ with angular advantage ($|\varphi_{ij}| < \pi/4$, $|q_{ij}| > 3\pi/4$), inflicting ΔH on the target. Collision and boundary penalties follow the hierarchical structure in (15), where immediate penalties are triggered at $d < 2$ m and graduated warnings are imposed within 5 m $< d <$ 15 m. Episodes terminate when all UAVs of either team are eliminated or the maximum of 400 timesteps is reached. Formation recommendations are governed by the numerical ratio r_{num} and spatial dispersion metrics d_{A}, d_{B} in (22)–(24).

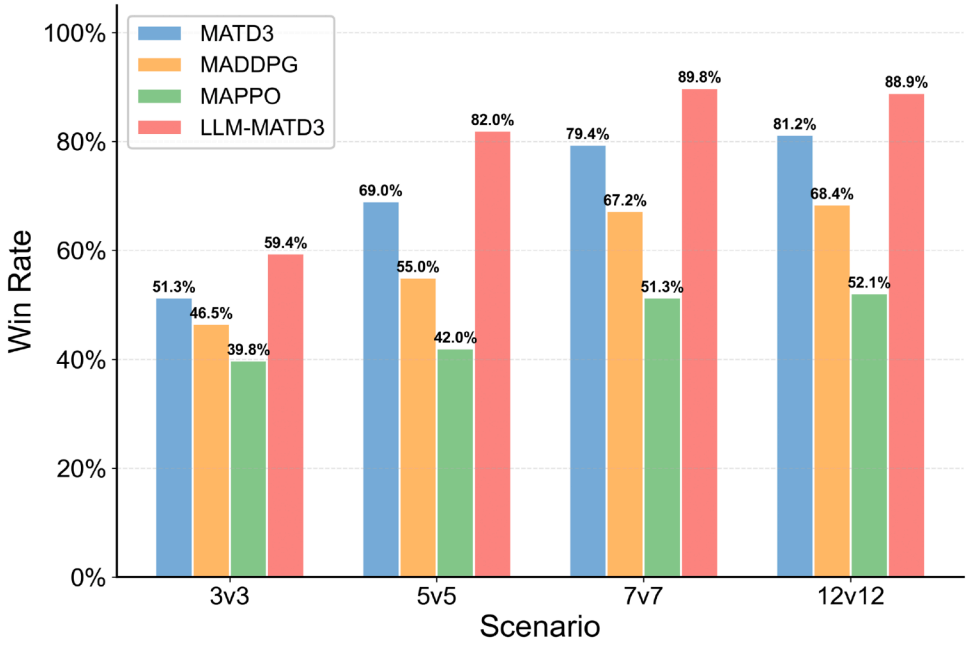


(a) Task success rates: LLM vs. Fixed assignment

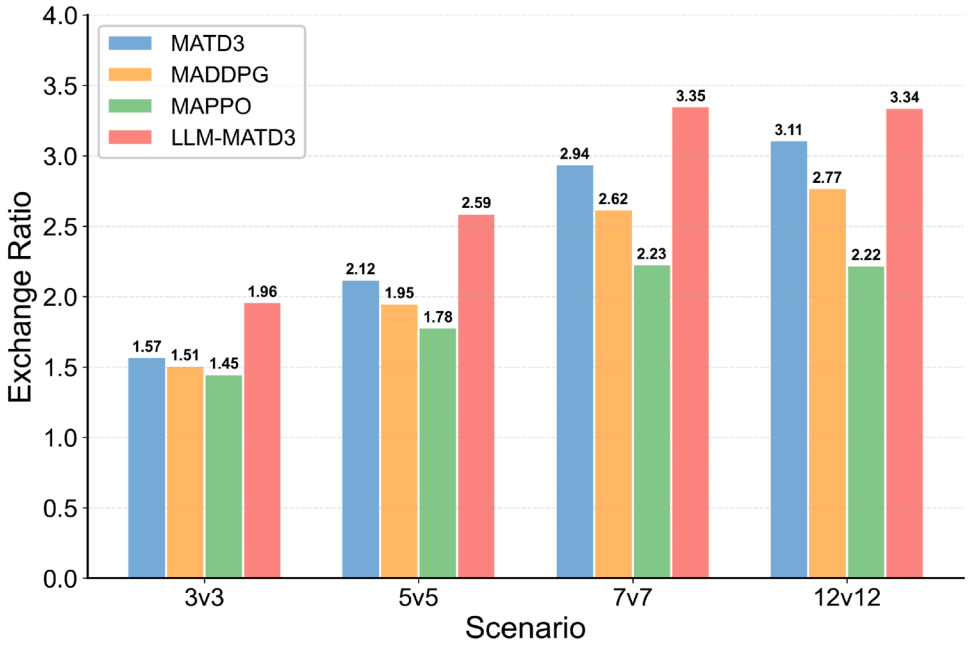


(b) System performance under different task allocation ratios

Fig. 8. Task assignment and allocation ratio analysis.



(a) Win rate comparison across different scale scenarios



(b) Exchange ratio comparison across different scale scenarios

Fig. 9. Scalability analysis across different game scenarios.

Table 1
UAV system parameters.

Component	Parameters	Values
UAV Physical Characteristics	Max Velocity v_{max}	45.0 m/s
	Mass m_{uav}	1.0 kg
Game Parameters	Initial Health H	100.0
	Task Radius ρ_i	300.0 m
Sensors	Number	11
	Detection Range l_{max}	200.0 m
Environment Configuration	Field Dimensions	1600×1600 m^2
	Obstacle Count	3–5
	Obstacle Radius	[30.0, 100.0] m

The proposed method is evaluated using three metrics: **(1) Win rate:** the proportion of episodes in which A-Team eliminates all B-Team UAVs; **(2) Exchange ratio:** the ratio of eliminated B-Team UAVs to lost A-Team UAVs, reflecting game efficiency; **(3) Cumulative reward:** the total reward accumulated by A-Team over an episode, indicating overall policy quality. In addition, the LLM component utilizes the DeepSeek-R1 model (14B parameters) as game analyzer, deployed locally via the Ollama framework and fine-tuned using Low-Rank Adaptation (LoRA). Specifically, the fine-tuning dataset comprises approximately 3000 input-output samples automatically generated from the simulation environment, covering game scenarios. During the training,

Table 2
Training hyperparameters.

Component	Parameters	Values
RL Parameters	Discount Factor γ	0.99
	Soft Update Rate τ_u	0.005
	Batch Size	512
	Actor Learning Rate	3×10^{-4}
	Critic Learning Rate	3×10^{-4}
	Policy Noise	0.1
	Noise Clip Range	0.25
	Policy Update Frequency	3
	Replay Buffer Size	3×10^{5}
LLM Integration	Task Update Interval	100 steps
	Strategy Update Interval	300 steps
	LLM Factor λ	0.25
	Task Embedding Dimension	32
Training Config	Training Episodes	35,000
	Max Steps/Episode	400
	Evaluation Episodes	50
	Save Interval	500 episodes

4-bit NF4 quantization is employed to mitigate memory constraints. The key LoRA hyperparameters are set as: rank $r = 16$, $\alpha = 32$, and dropout = 0.05, using the AdamW optimizer (learning rate 2×10^{-4}) for 3 epochs. Furthermore, the LLM task scheduler updates every 100 timesteps for task allocation to reduce computational overhead, while game recommendations update every 300 timesteps.

4.2. Testings and results

4.2.1. Overall performance comparison

Based on the simulation environment and parameter settings, we first trained 5 vs.5 multi-UAV game policy models using four algorithms: LLM-MATD3, MATD3 [28], MADDPG [42], and MAPPO [43]. Fig. 7(a) demonstrates the average reward variation curves during training. The horizontal axis represents training episodes ($\times 10$, totaling 3500 episodes), and the vertical axis represents average reward. The red, blue, orange, and green curves represent the performance indicators for four algorithms respectively.

Fig. 7(a) shows the average reward curves during training. During the initial phase, all four algorithms achieved relatively low rewards, indicating that effective game rules had not yet been established. As training progressed, LLM-MATD3 demonstrated faster convergence rate compared to the other methods. After convergence, LLM-MATD3 achieved the highest average reward of approximately 1,600, outperforming MATD3 (~1,300), MADDPG (~1,200), and MAPPO (~900).

Fig. 7(b) presents win rate comparisons. LLM-MATD3 achieved the highest win rate of 82%, while MATD3, MADDPG, and MAPPO achieved 69%, 55%, and 42%, respectively. These results demonstrate that LLM-

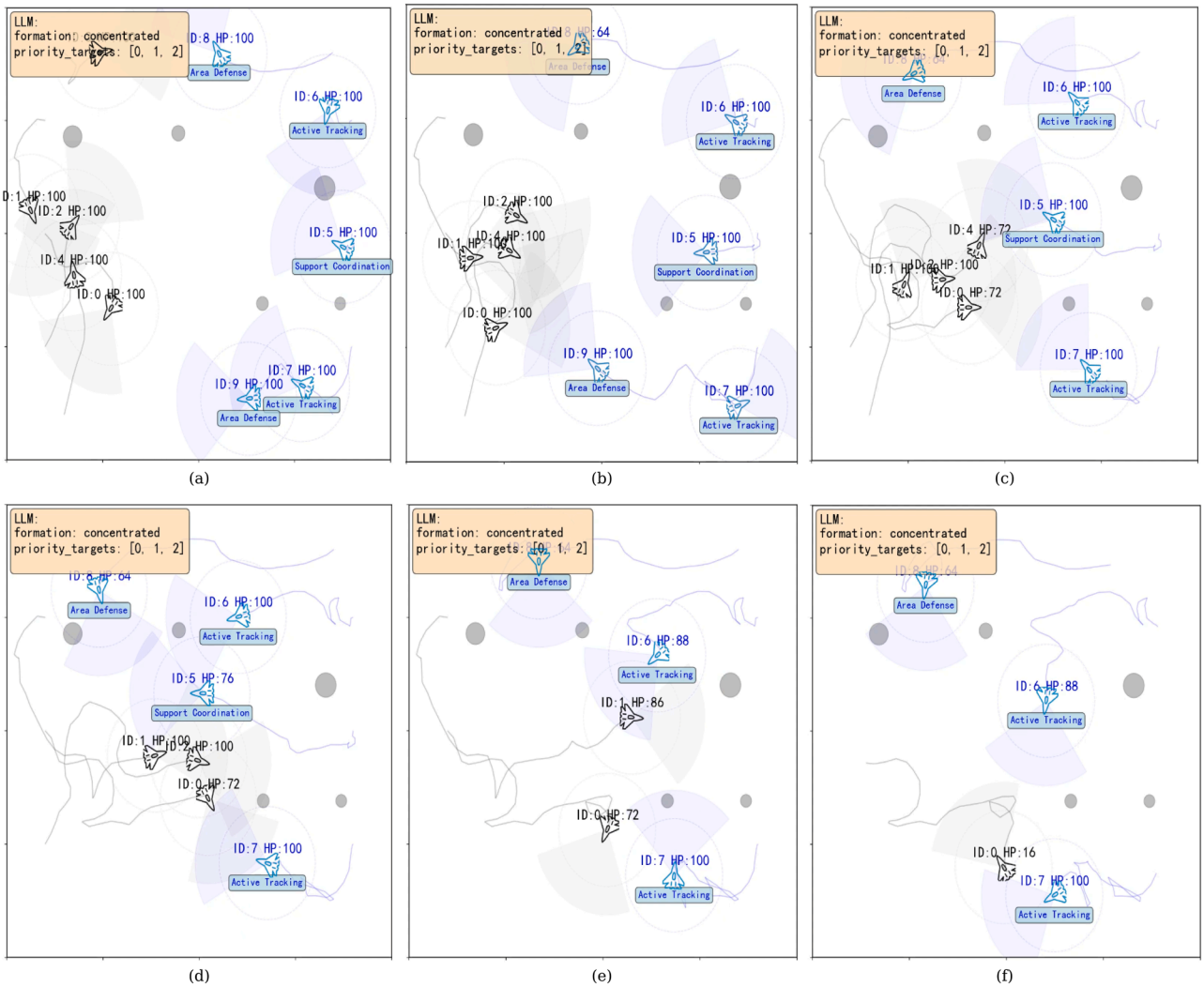


Fig. 10. 5-vs-5 game scenario-1 at different time steps. (a) $t = 0$. (b) $t = 80$. (c) $t = 160$. (d) $t = 240$. (e) $t = 300$. (f) $t = 400$.

MATD3 outperforms both traditional reinforcement learning algorithms and rule-based approaches, indicating that LLM-based game analysis effectively reduces the exploration space and improves policy quality.

4.2.2. Task allocation effectiveness

This section evaluates the effectiveness of LLM-based task allocation by comparing it with fixed allocation methods and analyzing the impact of allocation ratios. Fig. 8(a) presents execution success rates across three task types, with each data point representing the average of 100 independent simulation runs. LLM allocation outperforms fixed allocation in active tracking (80.5% vs. 63.2%) and cooperative support (46.4% vs. 31.6%) tasks. Notably, fixed allocation outperforms LLM allocation in area defense tasks (60.5% vs 48.7%), likely because predefined positions reduce coordination uncertainty. It should be noted that the area defense task is fundamentally a position-maintenance task where execution depends on local motion control rather than high-level reasoning, and fixed allocation provides more uniform spatial coverage compared to LLM's dynamic reallocation which may create temporary coverage gaps.

Fig. 8(b) compares win rates and exchange ratios across six task allocation configurations, where the ratio denotes Active Tracking : Area Defense : Support. Each configuration is evaluated over 100 Monte Carlo simulations. The results reveal a non-monotonic performance profile: the optimal 55:35:10 configuration achieves an 82.0% win rate and 2.59 exchange ratio while minimizing variance. Performance degrades when active tracking exceeds 70%, indicating that balanced task allocation is critical for overall effectiveness. Moreover, the box plot distributions indicate that the 55:35:10 configuration exhibits the smallest box height, reflecting superior stability across 100 simulations. Configurations with extreme ratios, such as 30:40:30 and 70:20:10, display wider boxes and greater variability. The isolated circle points beyond the box plots in Fig. 8(b) represent a few simulation runs with results notably different from the majority. The combination of highest mean performance and lowest variance confirms 55:35:10 as the most robust allocation results.

4.3. Different scale task scenarios

To evaluate the scalability and adaptability of the LLM-MATD3 framework across different-scale game environments, we conducted comparative experiments in different scale tasks. As shown in Fig. 9(a), in the 3v3 environment, LLM-MATD3 achieved a 59.4% win rate, outperforming MATD3 (51.3%), MADDPG (46.5%), and MAPPO (39.8%), though the margin is relatively modest. The exchange ratio data in Fig. 9(b) shows LLM-MATD3 reaching 1.96, exceeding MATD3 (1.57), MADDPG (1.51) and MAPPO(1.45). This indicates that in small-scale environments, agent coordination complexity remains relatively low, allowing traditional reinforcement learning methods to maintain reasonable effectiveness.

As the task scale expands to 7v7, decision space complexity increases significantly, and performance differentiation among algorithms



Fig. 11. 5-vs-5 game scenario-2 at different time steps. (a) $t = 0$. (b) $t = 80$. (c) $t = 160$. (d) $t = 240$. (e) $t = 300$. (f) $t = 400$.

becomes more pronounced. Fig. 9(a) shows LLM-MATD3 achieves a win rate of 89.8%, outperforming MATD3 (79.4%), MADDPG (67.2%), and MAPPO (51.3%) by 10.4, 22.6, and 38.5 percentage points respectively. In terms of exchange ratio Fig. 9(b), LLM-MATD3 achieves 3.35, outperforming MATD3 (2.94), MADDPG (2.62), and MAPPO (2.23).

To quantify performance reliability and validate statistical significance, independent experiments were conducted using 5 random seeds based on the final 50 evaluation episodes. The proposed LLM-MATD3 in this work consistently outperforms all three baselines with statistically improvements ($p < .05$ in 3v3; $p < .01$ in 5v5/7v7/12v12), achieving at least approximately 10%, 20%, and 35% higher win rates compared to MATD3, MADDPG, and MAPPO respectively. Moreover, LLM-MATD3 exhibits superior stability with lower standard deviations across all scales. It demonstrates that LLM-MATD3-based task allocation becomes increasingly effective in managing coordination challenges as the increasing number of agents in the game.

4.4. Visual analysis

To validate the practical effectiveness of LLM-MATD3, this section presents a visual analysis of two representative formation strategies in 5v5 game scenarios.

As shown in Fig. 10, The LLM recommends concentrated formation based on situational assessment. During the initial phase ($t = 0 - 80$), Team A UAVs converge into a tight coordination. In the mid-phase ($t = 160$–300), agents executing active tracking ($\tau = 0$) form a semicircular encirclement, constraining Team B to the central region. By the terminal phase ($t = 400$), Team A achieves complete elimination with minimal losses, demonstrating superior concentration and breakthrough capability.

As shown in Fig. 11, the LLM recommends a scattered formation strategy. Team A UAVs adopt dispersed initial positions to minimize concentrated damage. In the middle phase ($t = 160 - 300$), they engage from multiple angles, fragmenting Team B's defensive formation. Although UAV ID:7 gets damaged (Heath State:76), Team A maintains the advantage through flexible repositioning. The victory with only one damaged UAV demonstrates that scattered formations effectively balance multi-directional engagement with survivability.

In summary, concentrated formations provide superior game density, making them suitable for scenarios with clear force advantages or rapid breakthrough requirements. Scattered formations offer better spatial coverage and survivability against uncertain conditions. LLM-MATD3 dynamically selects formations based on real-time conditions, demonstrating adaptive decision-making in complex task environments.

5. Conclusion

This work proposes a Large Language Model-enhanced multi-agent deep reinforcement learning scheme(LLM-MATD3) for task decision-making. Through systematic simulation experiments, we demonstrate its effectiveness in multi-agent decision-making domains. The hierarchical task decomposition mechanism transforms abstract task labels into state-dependent sub-goal scores, effectively integrating LLM guidance with autonomous decision-making. This approach enhances collaborative capabilities in high-dimensional state spaces. Further analysis reveals that semantic enhancement and task decomposition mechanisms reduce decision space complexity, effectively addressing the curse of dimensionality problem. Additionally, dynamic task allocation enables agents to adjust execution modes based on game dynamics. In the future, we will further explore deep integration of LLM with reinforcement learning under game learning frameworks, hybrid dynamic-static task allocation strategies across diverse task types, multi-modal situational awareness with decision composition, and lightweight deployment solutions in continuous computing environments.

CRediT authorship contribution statement

Xinyu Dong: Writing – original draft, Validation, Methodology, Investigation; **Bo Li:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization; **Guangyu Zhang:** Writing – review & editing, Supervision, Investigation; **Bing Xiao:** Writing – review & editing, Supervision, Methodology; **Yuanshun Wang:** Validation, Investigation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] C. Zhang, G. Zhang, Q. Dong, Multi-variable finite-time observer-based adaptive-gain sliding mode control for fixed-wing UAV, IET Control Theory Appl. 15 (2) (2021) 223–247.
[2] B. Li, H. Liu, C.K. Ahn, W. Gong, Optimized intelligent tracking control for a quadrotor unmanned aerial vehicle with actuator failures, Aerosp. Sci. Technol. 144 (2024) 108803.
[3] S.-G. Kim, E. Lee, I.-P. Hong, J.-G. Yook, Review of intentional electromagnetic interference on UAV sensor modules and experimental study, Sensors 22 (6) (2022) 2384.
[4] F. Hu, T. Ma, X. Su, Adaptive fuzzy sliding-mode fixed-time control for quadrotor unmanned aerial vehicles with prescribed performance, IEEE Trans. Fuzzy Syst. 32 (7) (2024) 4109–4120.
[5] J. Wu, N. Zhang, D. Li, J. Bi, G. Han, A context-aware feature fusion method for multi-UAV cooperative air combat, IEEE Trans. Intell. Transp. Syst. 26 (5) (2025) 7197–7210.
[6] H. Liu, B. Li, C.K. Ahn, Asymptotically stable-learning-based formation control for multiquadrotor UAVs, IEEE Internet Things J. 12 (13) (2025) 24985–24995.
[7] Z. Zhen, Y. Chen, L. Wen, B. Han, An intelligent cooperative mission planning scheme of UAV swarm in uncertain dynamic environment, Aerosp. Sci. Technol. 100 (2020) 105826.
[8] J. Tang, X. Chen, X. Zhu, F. Zhu, Dynamic reallocation model of multiple unmanned aerial vehicle tasks in emergent adjustment scenarios, IEEE Trans. Aerosp. Electron. Syst. 59 (2) (2023) 1139–1155.
[9] H. Piao, Y. Han, H. Chen, X. Peng, S. Fan, Y. Sun, C. Liang, Z. Liu, Z. Sun, D. Zhou, Complex relationship graph abstraction for autonomous air combat collaboration: a learning and expert knowledge hybrid approach, Expert Syst. Appl. 215 (2023) 119285.
[10] Y. Cao, Y.-X. Kou, Z.-W. Li, A. Xu, Autonomous maneuver decision of UCAV air combat based on double deep q network algorithm and stochastic game theory, Int. J. Aerosp. Eng. 2023 (1) (2023) 3657814.
[11] Y. Li, W. Chen, S. Liu, G. Yang, F. He, Multi-UAV cooperative air combat target assignment method based on VNS-IBPSO in complex dynamic environment, Int. J. Aerosp. Eng. 2024 (1) (2024) 9980746.
[12] J.S. McGrew, J.P. How, B. Williams, N. Roy, Air-combat strategy using approximate dynamic programming, J. Guidance Control Dyn. 33 (5) (2010) 1641–1654.
[13] Z. Yang, D. Zhou, H. Piao, K. Zhang, W. Kong, Q. Pan, Evasive maneuver strategy for UCAV in beyond-visual-range air combat based on hierarchical multi-objective evolutionary algorithm, IEEE Access 8 (2020) 46605–46623.
[14] A. Xu, H. Li, Y. Hong, G. Liu, Autonomous decision-making for air gaming based on position weight-based particle Swarm optimization algorithm, Aerospace 11 (12) (2024) 1030.
[15] L. Wang, S. Zheng, S. Tai, H. Liu, T. Yue, UAV air combat autonomous trajectory planning method based on robust adversarial reinforcement learning, Aerosp. Sci. Technol. 153 (2024) 109402.
[16] X. Wang, Y. Wang, X. Su, L. Wang, C. Lu, H. Peng, J. Liu, Deep reinforcement learning-based air combat maneuver decision-making: literature review, implementation tutorial and future direction, Artif. Intell. Rev. 57 (1) (2024) 1.
[17] S. Zhang, L. Zhang, F. Xu, S. Cheng, W. Su, S. Wang, Dynamic deployment method based on double deep Q-network in UAV-assisted MEC systems, J. Cloud Comput. 12 (1) (2023) 130.

[18] L. Jia, C. Cai, X. Wang, Z. Ding, J. Xu, K. Wu, J. Liu, Multi-intent autonomous decision-making for air combat with deep reinforcement learning, Appl. Intell. 53 (23) (2023) 29076–29093.
[19] H. Zhang, Y. Wei, H. Zhou, C. Huang, Maneuver decision-making for autonomous air combat based on FRE-PPO, Appl. Sci. 12 (20) (2022) 10230.
[20] M.H. Lee, J. Moon, Deep reinforcement learning-based model-free path planning and collision avoidance for UAVs: a soft actor-critic with hindsight experience replay approach, ICT Express 9 (3) (2023) 403–408.
[21] H. Liu, B. Li, C.K. Ahn, Fixed-time reinforcement learning framework for containment coordination of networked aerial vehicles in inspection tasks, IEEE Trans. Veh. Technol. (2026), https://doi.org/10.1109/TVT.2026.3663248.
[22] W. Hu, Y. Wang, Q. Chen, P. Wang, E. Wu, Z. Guo, Z. Hou, TD3 agent-based nonlinear dynamic inverse control for fixed-wing UAV attitudes, IEEE Trans. Intell. Transp. Syst. 27 (2) (2026) 2607–2618.
[23] W. Du, S. Ding, A survey on multi-agent deep reinforcement learning: from the perspective of challenges and applications, Artif. Intell. Rev. 54 (5) (2021) 3215–3238.
[24] W. Guo, G. Liu, Z. Zhou, L. Wang, J. Wang, Enhancing the robustness of QMIX against state-adversarial attacks, Neurocomputing 572 (2024) 127191.
[25] H. Wang, X. Xie, L. Zhou, Transform networks for cooperative multi-agent deep reinforcement learning, Appl. Intell. 53 (8) (2023) 9261–9269.
[26] Z. Sun, H. Piao, Z. Yang, Y. Zhao, G. Zhan, D. Zhou, G. Meng, H. Chen, X. Chen, B. Qu, et al., Multi-agent hierarchical policy gradient for air combat tactics emergence via self-play, Eng. Appl. Artif. Intell. 98 (2021) 104112.
[27] K. Wan, D. Wu, B. Li, X. Gao, Z. Hu, D. Chen, ME-MADDPG: an efficient learning-based motion planning method for multiple agents in complex environments, Int. J. Intell. Syst. 37 (3) (2022) 2393–2427.
[28] Y. Zhou, X. Kong, K.-P. Lin, L. Liu, Novel task decomposed multi-agent twin delayed deep deterministic policy gradient algorithm for multi-UAV autonomous path planning, Knowl. Based Syst. 287 (2024) 111462.
[29] Y. Cao, H. Zhao, Y. Cheng, T. Shu, Y. Chen, G. Liu, G. Liang, J. Zhao, J. Yan, Y. Li, Survey on large language model-enhanced reinforcement learning: concept, taxonomy, and methods, IEEE Trans. Neural Netw. Learn. Syst. 36 (6) (2025) 9737–9757.
[30] L. Li, R. Tan, J. Fang, J. Xue, C. Lv, LLM-augmented hierarchical reinforcement learning for human-like decision-making of autonomous driving, Expert Syst. Appl. 294 (2025) 128736.
[31] G. Zhu, R. Zhou, W. Ji, S. Zhao, LAMARL: LLM-aided multi-agent reinforcement learning for cooperative policy generation, IEEE Rob. Autom. Lett. 10 (7) (2025) 7476–7483.
[32] H.-S. Liu, S. Kuroki, T. Kozuno, W.-F. Sun, C.-Y. Lee, Language-guided pattern formation for swarm robotics with multi-agent reinforcement learning, in: 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2024, pp. 8998–9005.
[33] F. Zhang, J. Li, Z. Li, A TD3-based multi-agent deep reinforcement learning method in mixed cooperation-competition environment, Neurocomputing 411 (2020) 206–215.
[34] Z. Li, R. Zhang, Z. Wang, Z. Xie, Y. Song, LLM-guided decision-making toolkit for multi-agent reinforcement learning, Neurocomputing 638 (2025) 130105.
[35] J. Han, Y. Zhu, J. Yang, A deep reinforcement learning method for collision avoidance with dense speed-constrained multi-UAV, IEEE Rob. Autom. Lett. 10 (3) (2025) 2152–2159.
[36] G. Zhu, R. Zhou, W. Ji, S. Zhao, Lamarl: LLM-aided multi-agent reinforcement learning for cooperative policy generation, IEEE Rob. Autom. Lett. 10 (7) (2025) 7476–7483.
[37] H.-S. Liu, S. Kuroki, T. Kozuno, W.-F. Sun, C.-Y. Lee, Language-guided pattern formation for swarm robotics with multi-agent reinforcement learning. In 2024 IEEE, in: RSJ International Conference on Intelligent Robots and Systems (IROS), 2024, pp. 8998–9005.
[38] J. Wen, Z. Li, M. Xi, J. He, A LLM-assisted AUV 3D path planning scheme under ocean current interference via reinforcement learning, IEEE Internet Things J. 12 (19) (2025) 39185–39196.
[39] D. Xu, E. Qiao, T. Gu, H. Fu, C. Sun, H. Guo, Y. Liu, LEAD: LLM-enhanced deep reinforcement learning for stable decision-making in critical autonomous driving scenarios, Neurocomputing (2025) 131619.
[40] C. Xu, Y. Chu, Q. Gao, Z. Wu, J. Wang, Y. Yue, W. Dominik, X. Zhu, Autonomous unmanned surface vehicle docking using large language model guide reinforcement learning, Ocean Eng. 323 (2025) 120608.
[41] H. Pang, Z. Wang, G. Li, Large language model guided deep reinforcement learning for safe autonomous vehicle decision making, Transp. Res. Part C Emerg. Technol. 184 (2026) 105511.
[42] R. Lowe, Y.I. Wu, A. Tamar, J. Harb, O. Pieter Abbeel, I. Mordatch, Multi-agent actor-critic for mixed cooperative-competitive environments, Adv. Neural Inf. Process. Syst. 30 (2017).
[43] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, Y. Wu, The surprising effectiveness of ppo in cooperative multi-agent games, Adv. Neural Inf. Process. Syst. 35 (2022) 24611–24624.